ASSIGNMENT-8.2

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Batch: 08

Lab 8: Test-Driven Development with AI – Generating and Working with Test Cases

Task Description

Task 1 – Test-Driven Development for Even/Odd Number Validator

- Use AI tools to first generate test cases for a function is_even(n) and then implement the function so that it satisfies all generated tests.

Requirements:

- Input must be an integer
- Handle zero, negative numbers, and large integers

Example Test Scenarios:

is_even(2) → True

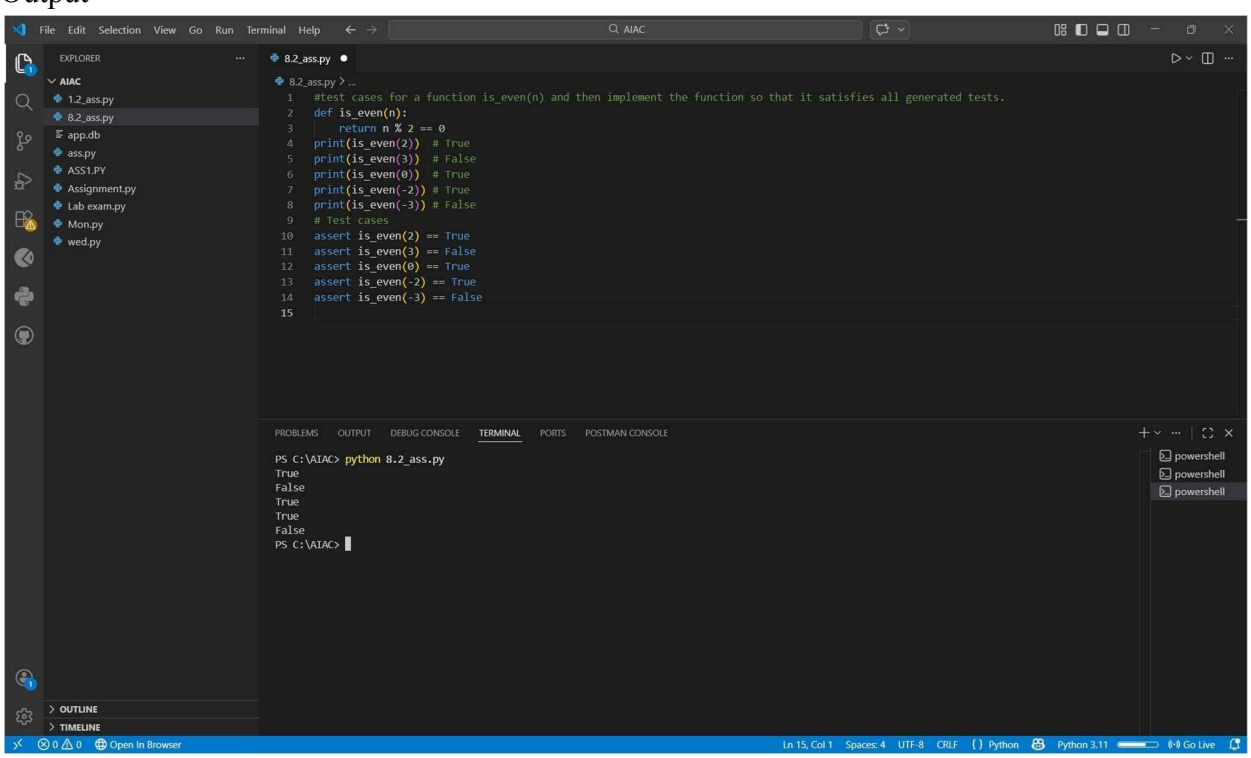
is_even(7) → False

is_even(0) → True

is_even(-4) → True

is_even(9) → False

Expected Output

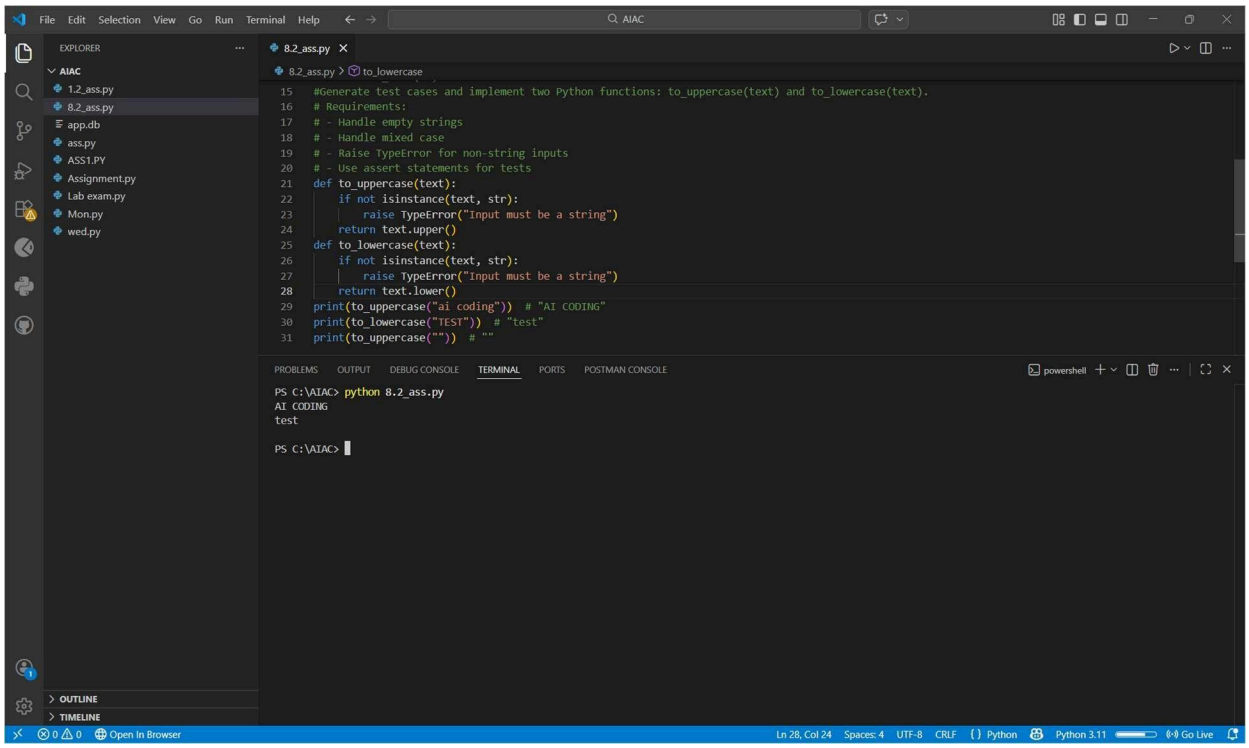


- A correctly implemented is_even() function that passes all AI- generated test cases

Task Description

Task 2 – Test-Driven Development for String Case Converter

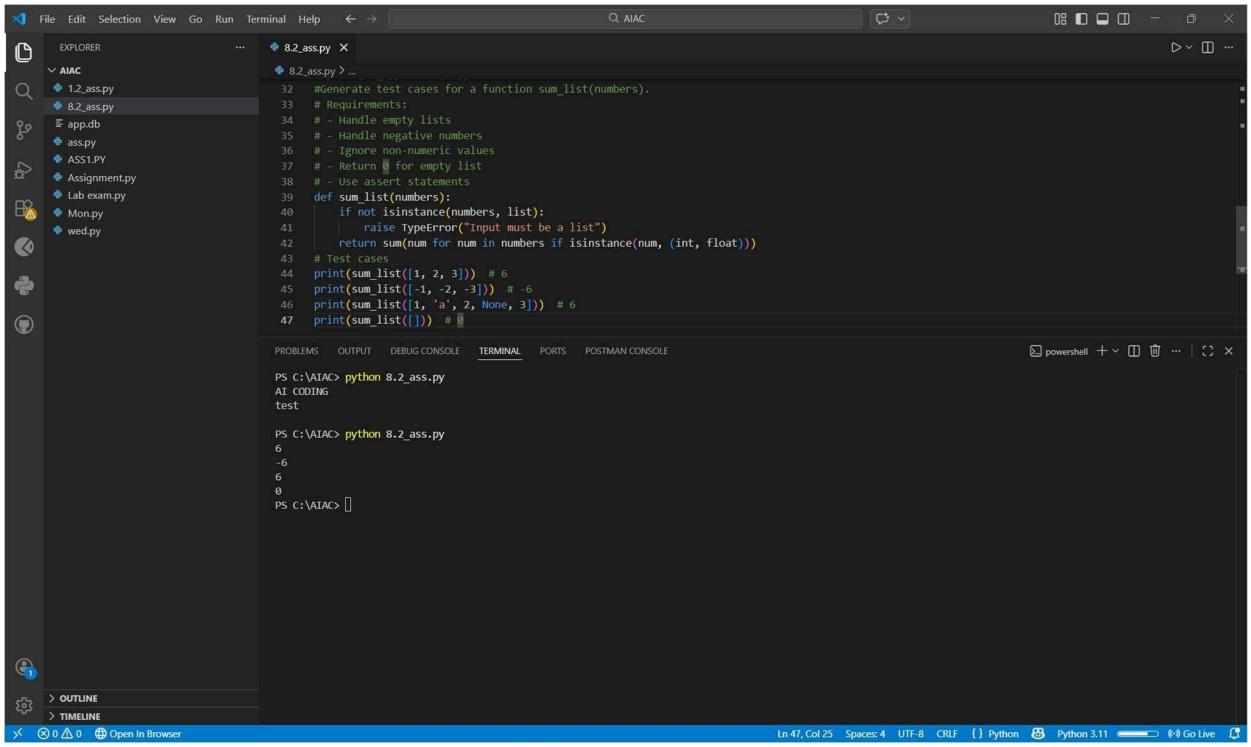
• Ask AI to generate test cases for two functions:
• to_uppercase(text)
• to_lowercas
e(text)
Requirements:
• Handle empty strings
• Handle mixed-case input
• Handle invalid inputs such as numbers or
None Example Test Scenarios:
to_uppercase("ai coding") → "AI CODING"
to_lowercase("TEST") → "test"
to_uppercase("") → ""
to_lowercase(None) → Error or safe handling
Expected Output



• Two string conversion functions that pass all
AI-generated test cases with safe input handling.
Task Description
Task 3 – Test-Driven Development for List Sum Calculator
• Use AI to generate test cases for a function
sum_list(numbers) that calculates the sum of list elements.
Requirements:
• Handle empty lists
• Handle negative numbers
• Ignore or safely handle non-numeric
values Example Test Scenarios:
sum_list([1, 2, 3]) → 6

sum_list([]) → 0
sum_list([-1, 5, -4]) → 0

sum_list([2, "a", 3]) → 5
Expected Output



- A robust list-sum function validated using

AI-generated test cases.
Task Description
Task 4 – Test Cases for Student Result Class

- Generate test cases for a StudentResult class with the

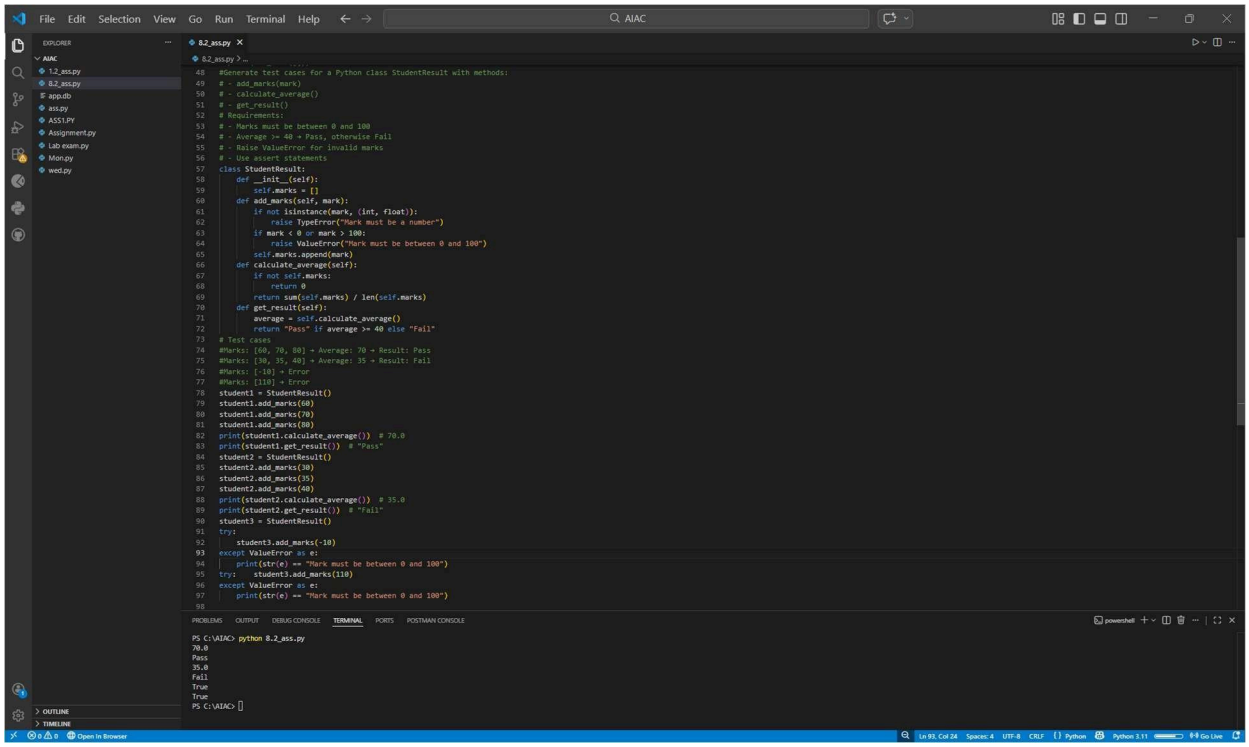
following methods:
• add_marks(mark)
• calculate_average()

- get_re

sult()
Requirements
:
• Marks must be between 0 and 100

- Average ≥ 40 → Pass,

otherwise Fail Example Test
Scenarios:
Marks: [60, 70, 80] → Average: 70 → Result: Pass
Marks: [30, 35, 40] → Average: 35 → Result: Fail
Marks: [-10] → Error
Expected Output



```python
#Generate test cases for a Python class StudentResult with methods:
# - add_marks(mark)
# - calculate_average()
# - get_result()
# Requirements:
# - Marks must be between 0 and 100
# - Average >= 40 → Pass, otherwise Fail
# - Raise ValueError for invalid marks
# - Use assert statements
class StudentResult:
    def __init__(self):
        self.marks = []
    def add_marks(self, mark):
        if not isinstance(mark, (int, float)):
            raise TypeError("Mark must be a number")
        if mark < 0 or mark > 100:
            raise ValueError("Mark must be between 0 and 100")
        self.marks.append(mark)
    def calculate_average(self):
        if not self.marks:
            return 0
        return sum(self.marks) / len(self.marks)
    def get_result(self):
        average = self.calculate_average()
        return "Pass" if average >= 40 else "Fail"
# Test cases
#Marks: [60, 70, 80] → Average: 70 → Result: Pass
#Marks: [30, 35, 40] → Average: 35 → Result: Fail
#Marks: [-10] → Error
#Marks: [110] → Error
student1 = StudentResult()
student1.add_marks(60)
student1.add_marks(70)
student1.add_marks(80)
print(student1.calculate_average())  # 70.0
print(student1.get_result())  # "Pass"
student2 = StudentResult()
student2.add_marks(30)
student2.add_marks(35)
student2.add_marks(40)
print(student2.calculate_average())  # 35.0
print(student2.get_result())  # "Fail"
student3 = StudentResult()
try:
    student3.add_marks(-10)
except ValueError as e:
    print(str(e) == "Mark must be between 0 and 100")
try:    student3.add_marks(110)
except ValueError as e:
    print(str(e) == "Mark must be between 0 and 100")
```

```
PS C:\AIAC> python 8.2_ass.py
70.0
Pass
35.0
Fail
True
True
PS C:\AIAC>
```

- A fully functional StudentResult class that passes all AI- generated test

Task Description

Task 5 – Test-Driven Development for Username Validator

Requirements:

- Minimum length: 5 characters
- No spaces allowed
- Only alphanumeric characters

Example Test Scenarios:

is_valid_username("user01") → True

is_valid_username("ai") → False

is_valid_username("user name") → False

is_valid_username("user@123") → False

Expected Output

File Edit Selection View Go Run Terminal Help AIAC

8.2_ass.py

8.2_ass.py > is_valid_username

```python
#Generate test cases for a function is_valid_username(username).
# Requirements:
# - Minimum length: 5 characters
# - No spaces allowed
# - Only alphanumeric characters
# - Return True or False
# - Use assert statements
def is_valid_username(username):
    if not isinstance(username, str):
        raise TypeEr
    if len(username)
        return False
    if ' ' in userna
        return False
    if not username.isalnum():
        return False
    return True
# Test cases
#is_valid_username("user01") → True
# is_valid_username("ai") → False
# is_valid_username("user name") → False
# is_valid_username("user@123") → False
print(is_valid_username("user01"))  # True
print(is_valid_username("ai"))  # False
print(is_valid_username("user name"))  # False
print(is_valid_username("user@123"))  # False
```

(method) def isalnum() -> bool

Return True if the string is an alpha-numeric string, False otherwise.

A string is alpha-numeric if all characters in the string are alpha-numeric and there is at least one character in the string.

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE powershell

```
PS C:\AIAC> python 8.2_ass.py
True
False
False
False
PS C:\AIAC>
```

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